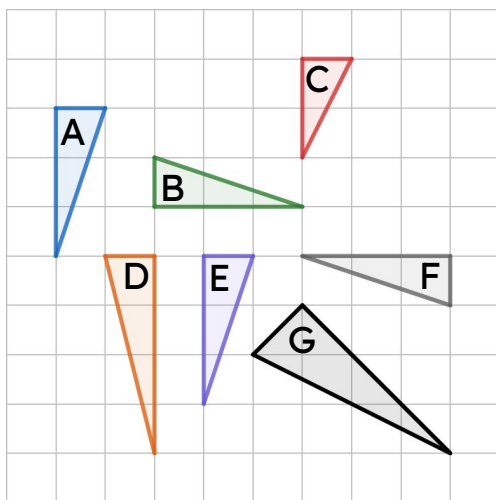




Checking understanding of basic transformations

Task A: Congruency

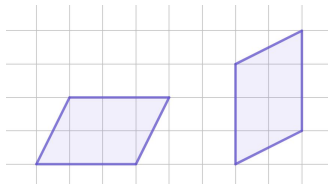
- 1) Which of these triangles are **congruent**?



A, B, E and F are **congruent** to each other.

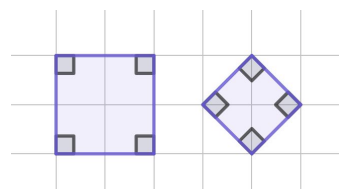
- 2) Are the two shapes **congruent** or not? If not, why not?

a)



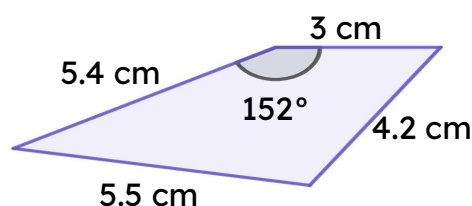
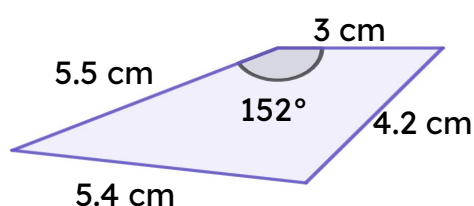
Congruent

b)



Not congruent as the sizes are different.

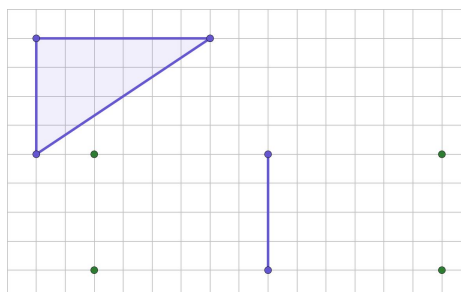
c)



These shapes are **not congruent** as the edges with the same length are in different places.

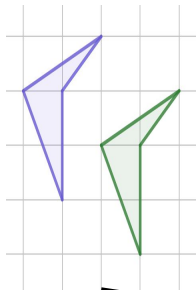
- 3) Mark where you could put the third vertex to make the two triangles **congruent**. One edge has been given.

There are four places you could have chosen.

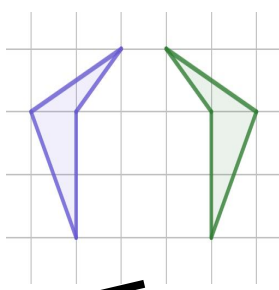


Task B: Describing changes

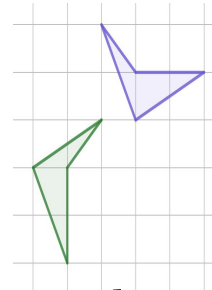
- 1) Match the descriptions to the correct figure.



The **image** is **congruent** but mirrored.



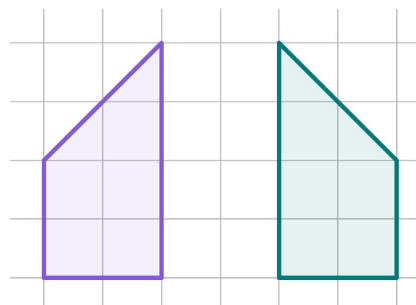
The **image** is **congruent** and has the same orientation as the **object**.



The **image** is **congruent** but the orientation has changed from the **object**.

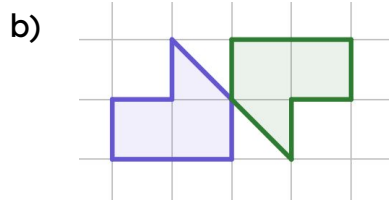
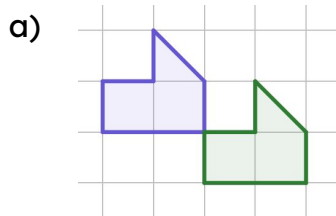
- 2) Complete the table regarding the properties of the image compared to the **object** and then use it to write a description.

Property	Changed - yes/no?
Orientation	<i>yes</i>
Lengths	<i>no</i>
Position	<i>yes</i>
Angles	<i>no</i>
Mirrored	<i>yes</i>



The lengths and angles are invariant. Its orientation and position has changed. It is a mirror image of the object.

- 3) Write a description for the two **transformations**.



Your descriptions may include:

- a) The **object** and **image** are **congruent** but in a different position.
 b) The **image** has a different orientation to the **object**.

